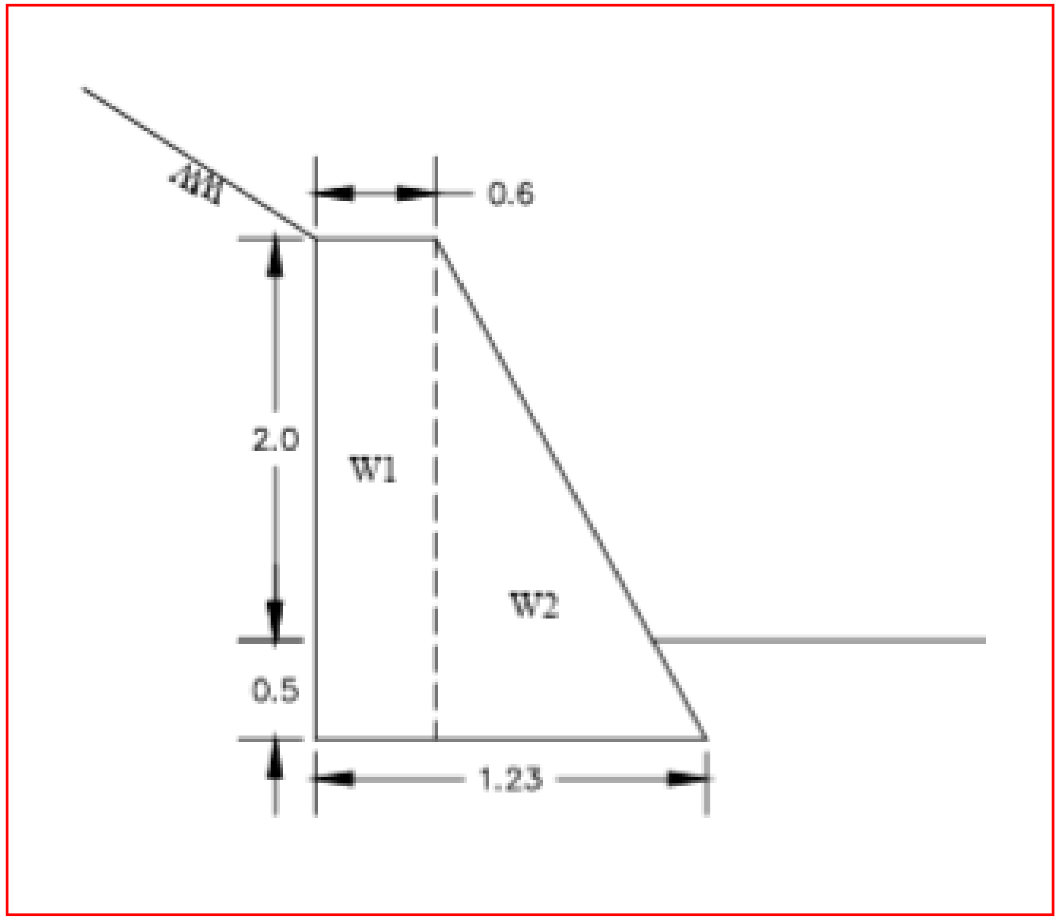
DESIGN OF RRM WALL 2.0m HT FOR AKHNOOR PACKAGE 7

INPUT DATA			
Unit wt. of RRM stone	γ_C	23.50	kN/m^3
Unit wt. of water	γ_W	9.81	kN/m^3
Unit wt. of soil(Saturated)	γ_{Sat}	21.0	kN/m^3
Angle of Backfill Slope	i	20.00	degrees
Angle of wall with Vertical	α	0.00	degrees
Angle of Friction between wall & earth fill	$\delta=2/3*\phi$	21.33	degrees
Length of Foundation Base	L	0.80	m
Total height	H	2.00	m
Horizontal co-eff. of acceleration,	$\alpha_h=\alpha_o*I*\beta$	0.150	g
Vertical co-eff. of acceleration,	$\alpha_v=\alpha_h*2/3$	0.100	g
Angle of internal Friction of soil	ϕ	32.0	deg.
SLIDING SURFACE PARAMETERS			
Cohesion, c		15.0	kPa
Angle of friction, f		32.0	deg
Factor Safety for Sliding (As per IS 14458 Part-2)			
Normal Case		1.5	
Earthquake Case		1.2	
Factor Safety for Over Turning (As per IS 14458 Part-2)			
Normal Case		2.0	
Earthquake Case		1.5	
Earth Pressure Co-efficients			
Co-efficient of Active Earth Pressure	$C_{a,\ Normal}$	0.374	
Co-efficient of Passive Earth Pressure	$C_{p,\ Normal}$	2.670	
Co-efficient of Active Earth Pressure	$C_{a,\ Earthquake}$	0.620	
Co-efficient of Passive Earth Pressure	$C_{p,\ Earthquake}$	1.613	



Self Weight Calculations:

Wall	Wt.	ex	ey	Wt.*ex	Wt.*ey
1	20.00	0.60	1.00	12.00	20.00
2	8.00	0.27	0.53	2.13	4.27
	28.00	**0.50**	**0.87**	**14.13**	**24.27**

EARTH PRESSURE COEFFICIENT CALCULATIONS

INPUT:

Unit weight of saturated soil	γ_s	=	21.00	kN/m^3
Angle of internal Friction of soil	ϕ	=	32.00	deg.
Inclination of Back Fill	i	=	20.00	deg.
Unit weight of RRM stone	γ_c	=	23.50	kN/m^3
Unit weight of water	γ_w	=	9.81	kN/m^3

Active Earth Pressure Coefficients: for Vertical wall face with back fill

(seismic coefficient method, IS 1893-2014 Part 3 & IS 1893-2016 Part 1)

Horizontal coefficient of acceleration,	$\alpha_h=\alpha_o*I*\beta$	α_h	=	0.000	g
Vertical Coefficient of acceleration	$\alpha_v=\alpha_h*2/3$	α_v	=	0.000	g
Angle of earth fill face with vertical		α	=	0.000	deg.
Angle of Friction between wall & earth fill	$\delta=2/3*\phi$	δ	=	21.33	deg.
		$\lambda(\pm)$	=	$\tan^{-1}\dfrac{\alpha_h}{1\pm\alpha_v}$	
		$\lambda(\pm)$	=	0.00	deg.
		0.6	=	0.00	deg.

Coefficient of active earth pressure, $C_a(\pm)$

(cl.8.1.1, IS 1893-1984, pg.46)

$$C_a(\pm) = \frac{(1\pm\alpha_v)\cos^2(\phi-\lambda-\alpha)}{\cos\lambda\cos^2\alpha\cos(\partial+\alpha+\lambda)} * \left[\frac{1}{1+\left\{\dfrac{\sin(\theta+\partial)\sin(\phi-i-\lambda)}{\cos(\alpha-i)\cos(\partial+\alpha+\lambda)}\right\}^{\frac{1}{2}}}\right]^2$$

	=	0.772	0.485	
$C_a(+)$		**0.374**	-------(1)	
	=	0.772	0.485	
$C_a(-)$		**0.374**	-------(2)	

Taking maximum of 1 & 2; $C_{a(Normal)}$ = **0.374**

Passive Earth Pressure Coefficients : for Vertical wall face with back fill

$C_{p(Normal)}$ = $1/C_{a(Normal)}$ $C_{p(Normal)}$ = 2.670

Active Earth Pressure Coefficients with earthquake: for Vertical wall face with back fill

(seismic coefficient method, IS1893-1984)

Horizontal coefficient of acceleration,	$\alpha_h=\alpha_o*I*\beta$	α_h	=	0.150	g
Vertical Coefficient of acceleration	$\alpha_v=\alpha_h*2/3$	α_v	=	0.100	g
Angle of earth fill face with vertical		α	=	0.000	deg.
Angle of Friction between wall & earth fill	$\delta=2/3*\phi$	δ	=	21.33	deg.
		$\lambda(\pm)$	=	$\tan^{-1}\dfrac{\alpha_h}{1\pm\alpha_v}$	
		$\lambda(+)$	=	7.77	deg.
		$\lambda(-)$	=	9.46	deg.

$$C_a(\pm) = \frac{(1\pm\alpha_v)\cos^2(\phi-\lambda-\alpha)}{\cos\lambda\cos^2\alpha\cos(\partial+\alpha+\lambda)} * \left[\frac{1}{1+\left\{\dfrac{\sin(\theta+\partial)\sin(\phi-i-\lambda)}{\cos(\alpha-i)\cos(\partial+\alpha+\lambda)}\right\}^{\frac{1}{2}}}\right]^2$$

	=	0.960	0.622	
$C_a(+)$		**0.597**	-------(3)	
	=	0.906	0.684	
$C_a(-)$		**0.620**	-------(4)	

Taking maximum of 3 & 4; $C_{a(Earthquake)}$ = **0.620**

Passive Earth Pressure Coefficients with earthquake: for Inclined wall face with back fill

$C_{p(Earthquake)}$ = $1/C_{a(Earthquake)}$ $C_{p(Earthquake)}$ = 1.613

AKHNOOR POONCH PACKAGE 7												
S. No.	Load Case Description	Loads	Vertical Force	Lever arm	Horizontal Force	Lever arm	Resisting Moments	Over turning Moments	Sum of Resisting moments	Sum of overturning moments	F.O.S. Calculated	F.O.S. Allowable
			V	LX	P	LY	MR	MO	∑MR	∑MO	MR / MO	As per IS 14458 (Part-2)
			kN/m	m	kN/m	m	kNm/m	kNm/m	kNm/m	kNm/m		
1	Case-1 (Normal Case)	Self weight	28.00	0.50			14.13		22	8	2.85	2.00
		Active soil pressure	10.39	0.80	11.81	0.67	8.31	7.87				
2	Case-2 (Normal Case with DBE)	Self weight	28.00	0.50			14.13		28	16	1.72	1.50
		Earthquake load (30% Vertical & Full Horizontal)	0.84	0.50	4.20	0.67		3.22				
		Active soil pressure	10.39	0.80	11.81	0.67	8.31	7.87				
		Active Soil pressure (dynamic increment)	6.81	0.80	7.74	0.67	5.44	5.16				

AKHNOOR POONCH PACKAGE 7

S. No.	Load Case Description	Loads	Vertical Force	Horizontal Force	Sum of Vertical Forces	Sum of Horizontal Forces	F.O.S. Calculated	F.O.S. Allowable (As per IS 14458 Part-2)
			V	P	∑V	∑P	$F = \frac{(\Sigma V)\tan\Phi + F_p}{\Sigma P}$	
			kN/m	kN/m	kN/m	kN/m		
1	Case-1 (Normal Case)	Self weight	28.00		38	12	2.41	>1.5
		Soil weight & Soil pressure	10.39	11.81				
		Passive pressure		4.49				
2	Case-2 (Normal Case with DBE)	Self weight	28.00		46	24	1.33	>1.2
		Earthquake load (30% Vertical & Full Horizontal)	0.84	4.20				
		Soil weight & Soil pressure	10.39	11.81				
		Soil pressure (dynamic increment)	6.81	7.74				
		Passive pressure		2.71				

AKHNOOR POONCH PACKAGE 7

S. No.	Load Case Description	Loads	Vertical Force	Lever arm	Horizontal Force	Lever arm	Moments	Sum of Moments	Sum of Vert. For.	Eccentricity (m)	Stresses at Ends (kN/m^2)	
			V	LX	P	LY	M	∑M	∑V	e=L/2-SM/SV	s = SV/L*(1±6e/L)	
			kN/m	m	kN/m	m	kNm/m	kNm/m	kNm/m		sA	sB
1	Case-1 (Normal Case)	Self weight	28.00	0.50			14.13	14.57	38.39	0.020	55.34	40.63
		Soil weight & Soil pressure	10.39	0.80	11.81	0.67	0.44					
2	Case-2 (Normal Case with DBE)	Self weight	28.00	0.50			14.13	12.48	46.03	0.13	113.16	1.92
		Earthquake load (30% Vertical & Full Horizontal)	0.84	0.50	4.20	0.67	-2.38					
		Soil weight & Soil pressure	10.39	0.80	11.81	0.67	0.44					
		Active Soil pressure (dynamic increment)	6.81	0.80	7.74	0.67	0.29					

PROJECT: BEARING CAPACITY CALCULATION FOR AKHNOOR POONCH PACKAGE 7

Safe Bearing Capacity Calculations

As per IS 6403-1981

Maximum Design Section		=	2	m
Load consider	DL	=	0	kN/m^2
	LL	=	24	kN/m^2
Block width	B	=	1.2	m
Eccentricity	e	=	0.02	m. (From Design sheet)
Effective width	B-2e	=	1.16	
Water Table depth from G. L.		=	20	m.
Water Table correction factor W'		=	1.00	
Pressure at base of wall=			55.24	kN/m^2 (From Design sheet)
Required q_{safe}		=	5.524	t/m2

Following are the shear parameter of the soil of the top layer

C	=	15	kN/m^2
ϕ	=	32	Degree
γ	=	21	kN/m3
D_f	=	0.4	m

For ϕ = 32 Degree

N_c	=	35.49
N_q	=	23.2
N_γ	=	30.2

Calculation of SBC

Depth Factor

$dc = 1 + 0.2(D_f/B)(N\phi)^{1/2}$ = 1.1899

$dq = d\gamma = 1$ for $\text{Ø} < 10^0$

$d_q = d_\gamma = 1 + 0.1(D_f/B)(N\phi)^{1/2}$ for $\text{Ø} > 10^0$ = 1.0949

Shape Factor

$Sc = Sq = S\gamma =$ 1 (For Strip Footing)

Inclination Factor

$i_c = i_q = i_\gamma =$ 1 (For vertical Wall)

$q_{Ult} = C N_c S_c d_c i_c + q N_q S_q d_q i_q + 0.5 \gamma B' N_\gamma S_\gamma d_\gamma i_\gamma W'$ **(As per IS 6403:1981)**

$= 350.62$ kN/m^2

$q_{safe} = 140.24$ kN/m^3 (FOS=2.5)

14.24t/m

Safe

2 **Depth of Foundation from Rankine's formula**

$$D_f = \frac{q_u}{\gamma}\left(\frac{1-\sin\varphi}{1+\sin\varphi}\right)^2$$

Df = 0.25 m ***Safe***